

Aditya Saini

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Systems Programming)

Expected Graduation: May 2028

Relevant Coursework: Linear Algebra, Multivariate Calculus, Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming

EXPERIENCE

Robotic Safety Research Assistant

January 2026 – Present

PurSec, Purdue Computer Science

West Lafayette, IN

- Researched safety of multimodal LLMs controlling embodied agents, formulating attack methods to create physical harm from benign individual steps for over 20 individual attacks spanning 8 different physical attack categories
- Created a dual-brain verification method to measure defensive success, consisting of RoboGuard LTL verification and the ASIMOV constitution checks running on GPT-4o to allow for quicker verifications for each attack
- Built a closed-loop harness through a state machine in Python that returns updated scene state and full action history to the verifier after every robot step, gating each proposed action on both safety brains before execution
- Built the Python simulation/eval pipeline using AI2-THOR rendering, Pillow + FFmpeg video output, and a Dockerized Ubuntu 22.04 + Xvfb headless harness for reproducible runs on machines with or without displays

Embedded Engineer

Sep. 2025 – Present

Purdue Electric Racing

West Lafayette, IN

- Designed and implemented the driver-facing dashboard UI for the Electric Racing competition vehicle on a Nextion HMI display using Nextion Editor, authoring all screen graphics for live telemetry, fault codes, and sensor data
- Wrote embedded C on an STM32F4 (ARM Cortex-M4) driving 6+ dashboard screens and button-mapped page transitions over UART, mapping CAN fault codes to prioritized driver alerts
- Rewrote the display command queue from fixed-width slots to a packed byte-level ring buffer, eliminating padding on the UART stream and fixing intermittent dropped commands on the Nextion dashboard display

Build President & Programming Captain

September 2021 – August 2025

FRC Team 2554 - The Warhawks

Edison, NJ

- Led 60+ students as Build President to District Championships after 18 years, earning 3 regional awards.
- Built a CUDA-accelerated OpenCV vision pipeline in Python for AprilTag detection, improving autonomous positioning accuracy by 66%, leading to a larger scoring volume with greater scoring efficiency.
- Fabricated a custom 30-button robot controller using a PSoC microcontroller, writing custom embedded C firmware for continuous input polling to ensure reliable driver input under competition conditions
- Diagnosed and resolved cross-system hardware failures across CAN bus-chained motor controllers and sensors
- Designed robot subsystems in Onshape through successive prototyping cycles, including the intake and drivetrain

PROJECTS

Apptum | TypeScript, Next.js, React, Supabase, OpenAI, KaTeX

July 2026 – Present

- Built an on-demand learning platform that generates a full course on any topic from a learner profile, with structured modules, long-form LaTeX notes, and MCQ practice flashcards per chapter
- Generated syllabi with OpenAI structured JSON schemas scaled to depth and duration, then bulk-wrote LaTeX notes and quizzes for every chapter so the entire course is ready up front
- Cached identical syllabi by SHA-256 profile hash, stored per-user courses in Supabase with RLS, and repaired JSON-corrupted LaTeX so chapter notes render math cleanly with KaTeX

Tava | TypeScript, React Native, Expo, Firebase, Python, OpenAI

September 2025

- Built a cross-platform mobile app with AI-powered search and swipe-based matching that recommends collaborators for student project ideas, ranking candidates by shared interests via LLM-assisted retrieval
- Shipped an interactive node-edge graph view that surfaces both direct and secondary connections between users, backed by Firebase authentication and data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML/CSS

Frameworks/Libraries: Next.js, React, React Native, Expo, OpenCV, KaTeX, Mapbox

AI/ML: OpenAI API, OpenRouter, GPT-4o, structured outputs, prompt engineering, LLM eval harnesses

Data & Tools: Supabase, Firebase, Docker, AWS EC2, Linux, Git/GitHub, CUDA, Cursor, Claude Code